

Activity: HOG for Pedestrian Detection

Introduction

In this activity, you'll implement the **Histogram of Oriented Gradients (HOG)** feature descriptor for pedestrian detection using Python.

HOG is a powerful technique in computer vision for detecting objects, particularly pedestrians, by capturing gradient orientations in localized image patches.

Learning Outcomes

By the end of this activity, you will:

1. Implement HOG feature extraction using Python and OpenCV.
2. Visualize HOG features to understand their representation of image structures.
3. Apply HOG for pedestrian detection in a sample image.

Working with Dev Container

To complete this assignment in a reliable and fully configured environment, please refer to the instructions in the file: `README-devcontainer.md`. This guide walks you through opening the assignment in **Visual Studio Code** using a **Dev Container**, which automatically installs all necessary Python libraries and tools. Following that setup ensures that the notebook runs smoothly without manual configuration or missing dependencies. Make sure to open the **main activity folder** in VS Code and follow the steps outlined in the Dev Container guide before starting the notebook.

Starter File

- Open the notebook: `activity_HOG_for_pedestrian_detection.ipynb` in the Code folder.
- You can run it locally or in Google Colab for convenience.
- Ensure `scikit-image`, `opencv-python`, `matplotlib`, `numpy`, and `imutils` are installed.
- The sample images (`image1.jpg` and `image2.png`) is located in the `assets` folder of the activity.

Activity Code Steps

This is a guided learning activity. You'll:

1. Load and preprocess the sample images (`image1.jpg` and `image2.png`) for HOG feature extraction.
2. Implement HOG feature extraction using OpenCV and `scikit-image`.
3. Visualize the HOG features to observe gradient orientations.
4. Apply HOG for pedestrian detection and interpret the results.

Conclusion

In this activity, you successfully:

- Loaded and preprocessed the sample image for HOG feature extraction.
- Implemented the **HOG** feature descriptor using Python.
- Visualized HOG features to understand their role in encoding image structures.
- Applied HOG for pedestrian detection, gaining insights into its effectiveness in computer vision tasks.